

Multiple Choice (10 marks)

1. Computational complexity of Gradient descent is,
 - (a) linear in D
 - (b) linear in N
 - (c) polynomial in D
 - (d) dependent on the number of iterations
2. The disadvantage of Grid search is
 - (a) It can not be applied to non-differentiable functions.
 - (b) It can not be applied to non-continuous functions.
 - (c) It is hard to implement.
 - (d) It runs reasonably slow for multiple linear regression.
3. Statement 1| The BLEU metric uses precision, while the ROGUE metric uses recall.
Statement 2| Hidden markov models were frequently used to model English sentences.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence
 $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 3
 - (b) 4
 - (c) 7
 - (d) 15
5. Which of the following are the spatial clustering algorithms?
 - (a) Partitioning based clustering
 - (b) K-means clustering
 - (c) Grid based clustering
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Lasso regression is more effective for feature selection due to its ability to shrink some coefficients to zero.
7. True or False: The model derived from best-subset selection is always identical to the final model after feature selection is completed.
8. True or False: The dimensionality of the null space of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is 2.
9. True or False: The joint probability distribution $P(X, Y, Z)$ for the Bayes net $X \leftarrow Y \rightarrow Z$ is given by $P(Y) * P(X|Y) * P(Z|Y)$.
10. True or False: The original ResNets and Transformers are both architectures that utilize feedforward neural networks for their operations.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to count occurrences of a character in a repeated string.
12. Write a python function to check whether the given number can be represented as difference of two squares or not.

End of Paper